IAA & IQA-Adapter

Knowledge Transfer from Image Quality Assessment to Diffusion Models

James Hsu – June 27th 2025

Research Motivation

- Diffusion models can generate high-fidelity images.
- However, they may lack **perceptual quality** consistency.
- Existing methods **do not consider image quality assessment (IQA)** during generation.
- Goal: Guide generation using **IQA knowledge**.

Image Aesthetic Assessment (IAA)

- Task: Predict human perception of **image beauty / attractiveness**
- Focus: *Composition*, *color harmony*, *emotion*, and *visual appeal*
- Compared to IQA: More **subjective** and **semantic**
- Ground truth: Usually human ratings from datasets like AVA

IAA vs IQA

Property	IQA (Image Quality)	IAA (Image Aesthetics)
Goal	Measure technical fidelity	Assess visual beauty
Type	Objective	Subjective
Metrics	PSNR, SSIM, BRISQUE	Human ratings, MOS (1–10)
Typical use	Compression, restoration	Photo filtering, AI art

IAA Pipeline (Simplified)

1. **Input image**
2. **Feature extraction** with CNN or Transformer
3. **Prediction**:
   - Regression → Score (1–10)
   - Classification → High/Low Aesthetic
   - Ranking → Compare image pairs
4. **Output** aesthetic score or label

Popular Datasets

- **AVA (Aesthetic Visual Assessment)** (link)

  > 250K+ images with 1–10 ratings from amateur photographers

- **AADB (Aesthetics and Attributes Database)** (link)

  > Includes both scores and visual attributes (symmetry, color balance)

- **CUHK-PQ** (link)

  > Binary labels: High aesthetic vs Low aesthetic

Models for IAA

- **NIMA (Neural Image Assessment)** – Google (paper)

  > Outputs aesthetic score distribution (1–10)

- **AesRankNet** (paper)

  > Uses ranking loss to compare image pairs

- **Transformer-based models**

  > Capture global aesthetics via attention

Learning Strategies

Approach	Description	Pros	Cons
Regression	Predicts continuous scores	Fine-grained output	Needs many labels
Classification	High vs Low aesthetic label	Simpler task	Less detailed
Ranking	Pairwise preference learning	Human-like behavior	Needs image pairs

Applications

- **Social media**: Rank or filter beautiful photos
- **Smart photography**: Auto-select best shot
- **AI generation**: Filter high-aesthetic results
- **Design / Ad image selection**
- **Camera feedback**: Real-time composition suggestions

IAA in Generative Models

- Combine with diffusion or GAN models
- Filter high-aesthetic generations
- Use in reward models (e.g. ImageReward, Pick-a-Pic)
- Trend: Not just realism, but *visual preference*

Summary

- IAA ≠ IQA: It’s about beauty, not clarity
- Powered by deep learning and human-labeled datasets
- Emerging role in **creative AI** (photography, generative art)
- Essential for subjective image understanding

What is IQA-Adapter?

- IQA-Adapter: Exploring Knowledge Transfer from Image Quality Assessment to Diffusion-based Generative Models (paper)
- A novel adapter module for diffusion-based generative models.
- Injects **IQA model knowledge** into the image generation pipeline.
- Enables **perceptual feedback** during generation steps.

Key Techniques

- Integrate a **frozen IQA model** (e.g., for perceptual quality analysis).
- Use **multi-task learning**:
  - Generation loss
  - IQA-guided perceptual loss
- **Adapter design** allows flexible plug-in to various diffusion steps.

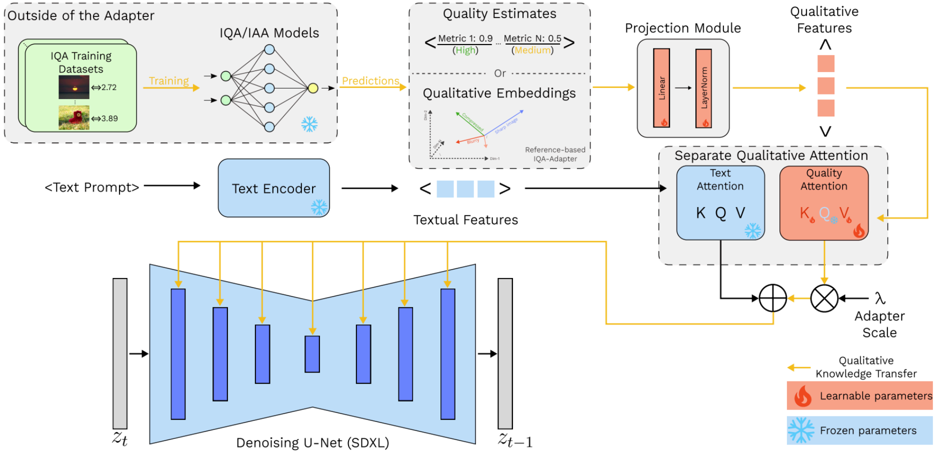


Figure 2. Overall architecture of the proposed **IQA-Adapter**. Yellow arrows depict IQA/IAA knowledge flow into the diffusion-based generator.

Experiments

- Base model:
  - SDXL
- Training dataset:
  - IQA-Adapters were trained on the CC3M (link) dataset (~3 million images) for 24,000 steps, followed by fine-tuning on a subset of the LAION-5B dataset (~170k images with an aesthetics score > 6.5) for an additional 3,000 steps
- Evaluation metrics:
  - GenEval (link)
  - FID, IS
  - CLIP

Results Summary

- GenEval

Models in IQA-Adapter	Two Object↑	Attribute Binding↑	Colors↑	Counting↑	Single Object↑	Position↑	Overall↑
MANIQA (PIPAL)	73.23%	20.25%	86.17%	36.56%	96.56%	10.50%	53.88%
TOPIQ (KONIQ)	71.97%	18.75%	85.11%	38.75%	98.12%	13.75%	54.41%
CLIPIQA+, LIQE-MIX	71.72%	20.25%	85.64%	41.25%	97.81%	11.75%	54.74%
TOPIQ, LAION-AES	69.70%	18.75%	85.90%	45.31%	99.38%	13.00%	55.34%
3xARNIQA,LIQE-MIX (different datasets)	**73.99%**	19.25%	**89.36%**	39.69%	**99.69%**	13.75%	55.95%
CLIP-IQA+	72.73%	22.75%	88.03%	43.44%	98.44%	12.25%	56.27%
HYPER-IQA	73.99%	**25.25%**	85.90%	39.69%	98.75%	**14.75%**	56.39%
TOPIQ (FLIVE)	72.73%	21.75%	87.77%	**45.94%**	99.38%	13.00%	**56.76%**
Base Model	73.74%	21.75%	88.30%	43.75%	99.69%	10.50%	56.29%

Table 1. Results of the IQA-Adapters trained with different IQA/IAA models on GenEval benchmark. Percents represent the accuracy of object-detection model on generated images. Results for all evaluated adapters are available in supplementary Table 6.

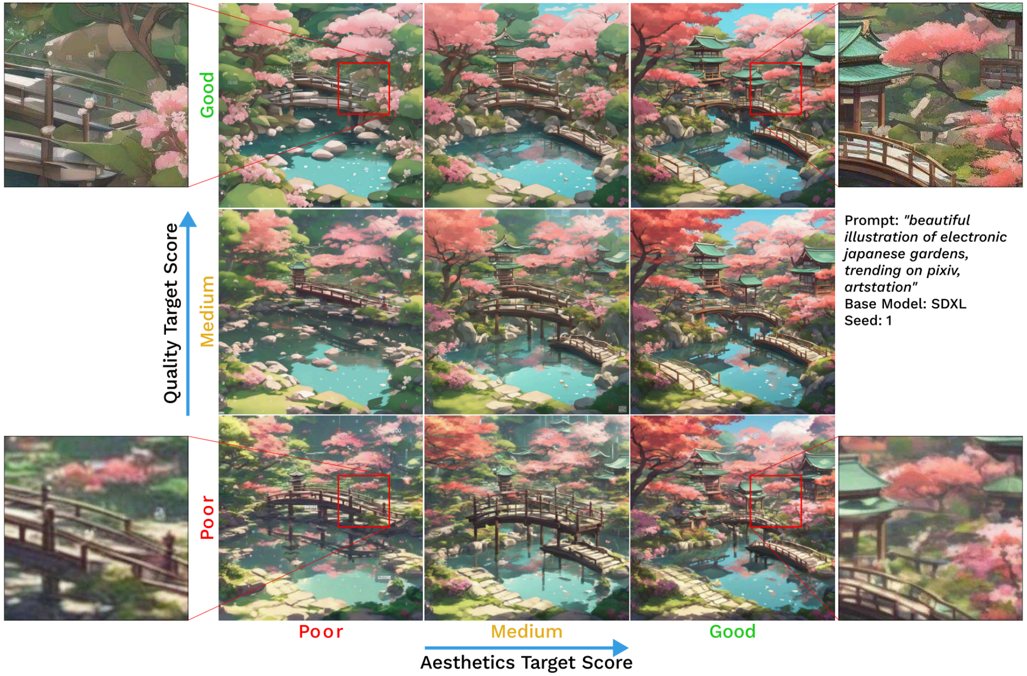
Results Summary

- FID, IS, CLIP

Models in IQA-Adapter	FID↓ Full	FID↓ (Top-25%)	FID↓ (Top-10%)	IS↑	CLIP-T↑	CLIP-I↑
LAION-AES	23.94	28.96	34.53	34.27±0.85	26.73	69.75
MUSIQ(KONIQ), MUSIQ(AVA)	22.48	24.96	29.68	37.00±1.43	26.79	69.47
NIMA (AVA)	22.32	25.65	30.55	37.72±1.08	26.70	69.80
TRES (FLIVE)	22.27	22.82	27.21	37.90±0.76	26.50	69.52
TOPIQ (AVA)	22.25	25.50	30.40	36.86±0.94	26.78	69.83
ARNIQA (3 versions), LIQE-MIX	21.95	22.92	27.58	37.55±1.02	26.69	69.62
TOPIQ (4 versions)	21.93	23.69	28.32	36.99±1.76	26.79	69.74
MANIQA (KONIQ)	21.74	23.85	28.57	37.63±1.23	26.91	69.61
CLIPIQA+, LIQE-MIX	21.43	22.45	27.02	38.33±1.83	26.70	69.65
TOPIQ, LAION-AES	21.36	23.53	28.44	36.89±1.33	26.83	70.02
MUSIQ (AVA)	21.20	24.92	30.08	36.42±1.39	26.93	69.96
ARNIQA (KONIQ)	21.13	22.70	27.53	37.32±0.87	26.86	69.53
TOPIQ (FLIVE)	21.04	**21.63**	**26.28**	37.93±0.70	26.64	69.54
HYPER-IQA	21.00	22.82	27.69	37.99±1.19	26.90	69.26
DBCNN	20.85	22.43	27.20	38.28±1.44	26.84	69.60
MUSIQ (KONIQ)	20.77	22.38	27.08	38.57±1.12	26.80	69.55
LIQE	20.76	22.34	27.21	37.72±1.46	26.82	69.81
CLIP-IQA+	20.45	21.89	26.55	37.66±1.05	26.80	**70.05**
ARNIQA (FLIVE)	20.44	21.75	26.58	38.25±1.20	26.85	69.99
ARNIQA (KADID)	20.35	22.50	27.56	37.67±1.31	26.76	69.32
LIQE-MIX	20.35	22.26	27.18	38.09±1.02	26.79	69.65
TOPIQ (SPAQ)	20.28	22.85	27.79	37.07±1.12	26.84	69.26
TOPIQ (KONIQ)	20.17	21.95	26.90	37.29±1.15	26.96	69.49
TOPIQ, HPSv2	19.67	22.08	27.40	36.71±1.45	27.00	69.12
CNNIQA	19.61	22.40	27.53	37.87±1.18	26.94	69.31
MANIQA (PIPAL)	**19.27**	21.88	27.19	37.98±1.46	26.77	69.48
Base Model	19.92	23.15	28.41	**39.44±1.66**	26.70	69.35
BeautifulPrompt	30.92	35.64	40.83	33.30±1.12	21.23	58.01
DiffusionDPO	29.57	34.04	38.88	36.93±1.04	**27.10**	68.74
Prompt Weighting	24.14	26.02	30.50	38.44±2.15	26.42	68.78
Q-Refine	20.29	23.41	28.56	39.05±1.11	26.83	69.11

Qualitative Improvements

- Sharper edges
- Fewer artifacts
- Better alignment to input prompts



Key Contributions

Item	Description
IQA-Guided Generation	Introduced adapter to guide generation via perceptual scores
Knowledge Transfer	IQA model knowledge transferred into generative process

Future Work

- Use stronger perceptual metrics (e.g., human preference models)
- Improve adapter design and inference efficiency

Summary

- IQA-Adapter bridges the gap between **quality assessment** and **image generation**.
- It demonstrates a **new paradigm**: using IQA as **guidance, not just evaluation**.

Thank You